Experiment 3: Automating Model Training using Jenkins with GCP Vertex AI

AIM

To study the automation of Machine Learning model training using Jenkins and its integration with GCP Vertex AI.

OBJECTIVES

1. To understand the role of Jenkins in automating ML workflows.
2. To study offline setup using Jenkins local jobs with scripts.
3. To understand how Jenkins triggers model training on Vertex AI.

THEORY

1. Automation in ML using Jenkins

Jenkins is an open-source automation server used for CI/CD. In ML projects, it can automatically trigger model training and testing when code changes occur.

Automation reduces manual effort, improves consistency, and speeds up experimentation.

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2. Offline Setup: Jenkins Local Job

In the offline setup, Jenkins is installed locally and configured to run shell or Python scripts for data processing and model training.

Models may be built using libraries like Scikit-learn, TensorFlow/Keras, or PyTorch. Jenkins ensures each run follows the same defined steps, maintaining reproducibility.

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3. Online Setup: Triggering Vertex AI

Google Cloud Platform provides

Vertex AI for scalable ML training.

Jenkins can trigger Vertex AI pipelines to execute training in the cloud, enabling higher compute power and centralized management.

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4. Integrated Workflow

In practice, scripts are tested locally using Jenkins, and then jobs trigger Vertex AI for large-scale cloud training.

This integration creates an automated, scalable, and structured ML pipeline from development to deployment.

CONCLUSION

In this experiment, we studied how Jenkins automates ML model training and integrates with Vertex AI for cloud execution.

This approach ensures efficiency, reproducibility, and scalability in ML workflows.